

Portfolio Case Study — Sales Performance Dashboard

1. Business Problem

Businesses often face challenges in monitoring sales performance, profitability, customer behavior, and regional growth effectively. Manual reporting makes it difficult to identify trends and make timely decisions.

The objective of this project was to build an interactive Power BI dashboard that provides real-time insights into:

- Sales performance
- Profitability
- Customer segments
- Regional contribution
- Monthly growth trends

2. Dataset Overview

The dataset contains sales transaction records with customer, product, and order-related information.

Main Tables Used

- Order Dataset
- Customer Dataset
- Product Dataset
- Calendar Table

Key Columns

- Order Date
- Sales
- Profit
- Quantity
- Region
- Segment
- Category
- Customer Name
- Product ID

Dataset Purpose

The dataset was used to analyze:

- Sales trends
- Profit margins
- Customer contribution
- Regional performance
- Product category profitability

3. Data Cleaning Summary

Data cleaning and transformation were performed using Power Query Editor.

Cleaning Steps Performed

- Removed duplicate records
- Checked missing/null values
- Corrected data types
- Created calculated date fields
- Standardized category and segment values
- Built a Calendar table for time intelligence analysis

Additional Enhancements

- Added Month Number for sorting
- Created Weekday and Week Number columns
- Improved column naming consistency

4. Data Modeling Diagram

A Star Schema model was implemented for efficient reporting and analysis.

Fact Table

- order

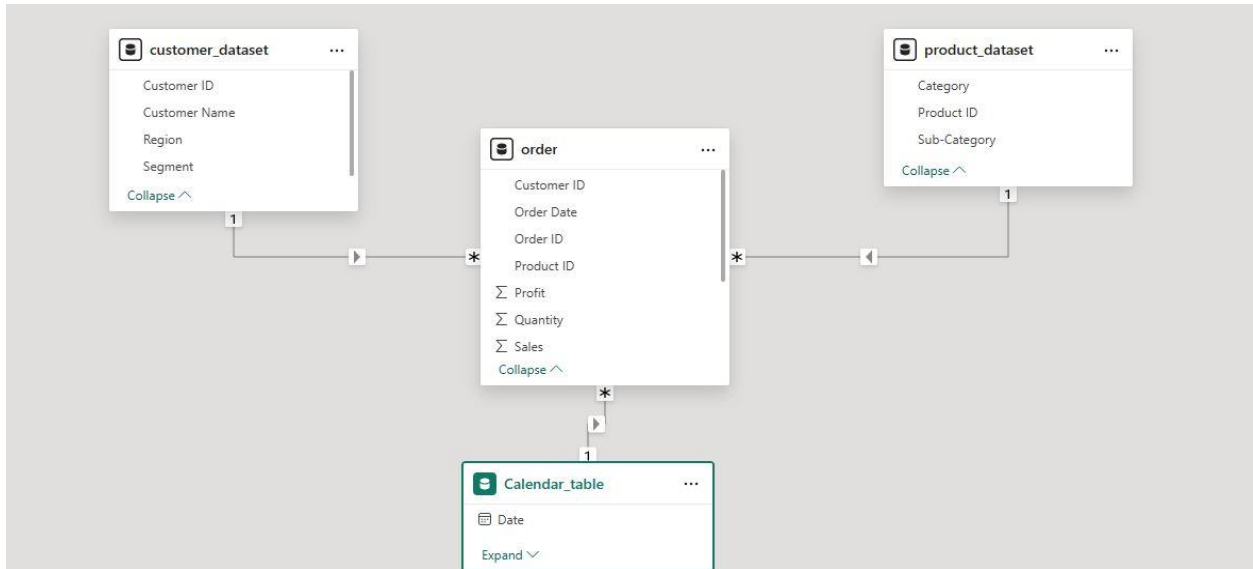
Dimension Tables

- Customer_Dataset
- Product_Dataset
- Calendar_Table

Relationships

- Customer ID → linked customer data
- Product ID → linked product data
- Date → linked calendar table

The model supports accurate filtering, aggregation, and time-based analysis.



5. KPIs & DAX Measures

Key KPIs

- Total Sales
- Total Profit
- Profit Margin
- Growth %
- Total Orders

Important DAX Measures

Total Sales :

$$Total\ Sales = SUM(order[Sales])$$

Total Profit :

Total Profit = SUM(order[Profit])

Total Orders :

Total Order = COUNT(order[Order ID])

Profit Margin:

*Profit Margin =
DIVIDE([Total Profit],[Total Sales])*

Month Moving Average Trend

```
3-Month Trend =  
AVERAGEX(  
    DATESINPERIOD(  
        Calendar_table[Date],  
        MAX(Calendar_table[Date]),  
        -3,  
        MONTH  
    ),  
    [Total Sales]  
)
```

YoY Growth %

```
YoY Growth % =  
VAR PreviousYearSales =  
    CALCULATE(  
        [Total Sales],  
        SAMEPERIODLASTYEAR(Calendar_table[Date])  
    )  
  
RETURN  
DIVIDE(  
    [Total Sales] - PreviousYearSales,  
    PreviousYearSales  
)
```

Customer Segment

```
Customer Segment =  
SWITCH(  
    TRUE(),  
    [Total Profit] > 5000, "High Value",  
    [Total Profit] > 2000, "Medium Value",  
    "Low Value"  
)
```

Dynamic Top N Filtering

```
Top N Sales =  
IF(  
    [Customer Rank] <= 10,  
    [Total Sales]  
)
```

Time Intelligence

Previous Month Sales:

```
«» DAX  
Previous Month Sales =  
CALCULATE(  
    [Total Sales],  
    PREVIOUSMONTH(Calendar[Date])  
)
```

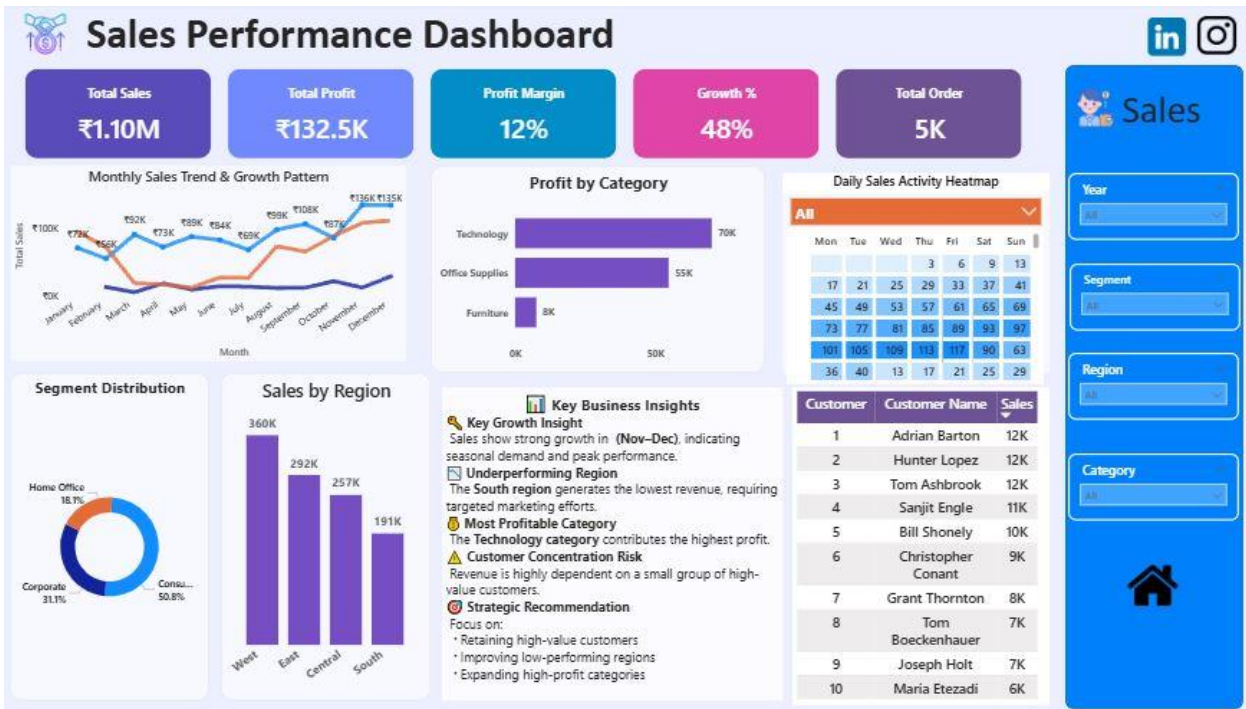
Growth %:

```
«» DAX  
Growth % =  
DIVIDE(  
    [Total Sales] - [Previous Month Sales],  
    [Previous Month Sales]  
)
```

Category Contribution

```
Category Contribution % =  
DIVIDE(  
    [Total Sales],  
    CALCULATE([Total Sales], ALL(Products[Category]))  
)
```

6. 🖥️ Dashboard Screenshots



📌 Dashboard Features

- KPI Cards
- Monthly Sales Trend
- Profit by Category
- Daily Sales Activity Heatmap
- Segment Distribution
- Regional Sales Analysis
- Customer Ranking Table
- Interactive Filters

📌 Interactive Elements

- Slicers for:
 - Year
 - Segment
 - Region
 - Category
- Home button for dashboard reset

7. Business Insights

Sales Growth Insight

Sales showed strong growth during (November–December), indicating seasonal demand and peak business performance.

Regional Analysis

- West region generated highest sales revenue.
- South region recorded the lowest performance and requires business attention.

Product Performance

Technology category contributed the highest profit compared to other categories.

Customer Analysis

A small group of high-value customers contributes a significant portion of total revenue, creating customer concentration risk.

Segment Analysis

Consumer segment accounted for the highest share of overall sales.

8. Strategic Recommendations

Improve Low-Performing Regions

Increase marketing campaigns and promotional activities in the South region.

Focus on High-Profit Categories

Expand investment in Technology products to maximize profitability.

Customer Retention

Develop loyalty programs for high-value customers to reduce concentration risk.

Business Expansion

Increase focus on Corporate segment growth opportunities.

Seasonal Planning

Prepare inventory and campaigns before to capitalize on peak demand periods.

9. 🛠️ Tools Used

📌 Visualization & Analytics

- Power BI

📌 Data Transformation

- Power Query Editor

📌 Data Modeling & Calculations

- DAX (Data Analysis Expressions)

📌 Data Source

- Excel/CSV Dataset

10. 📖 Learning Outcomes

Through this project, I improved my skills in:

- Data visualization
- Dashboard design
- DAX calculations
- Time intelligence functions
- Data modeling
- Business insight generation
- Interactive reporting in Power BI

This project enhanced my understanding of transforming raw business data into actionable insights for decision-making.